Aleksei Nemtyrev

Econometrics · Macroeconomics · Data Science

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Profile

Quantitative researcher and PhD candidate in econometrics (expected 2027), specialising in time-series modelling, causal inference and macroeconomics. My work separates signal from noise: I develop estimators with lower variance and test how reliably they measure effects. I handle the whole data workflow, from raw data to results people can act on. Before my PhD, I ran a retail company and took it from break-even to profit within a year.

Experience

PhD Candidate Aug 2023 – present
Tilburg University, Econometrics & Operations Research Tilburg, NL

- Develop statistical estimators, test bias and confidence-interval coverage, and release reproducible code.
- Run 20M+ bootstrap replications per study on an HPC cluster (MATLAB Parallel Computing Toolbox).
- Assemble datasets from FRED and the Quarterly Financial Report: own extraction code, cleaning, timing alignment.
- Build and estimate macroeconomic (DSGE) models to analyse monetary policy and test model restrictions.
- Teach *Data Science Methods* to 120 MSc students a year (in R: PCA, ridge/lasso, trees, double ML); designed my own model-selection and PCA assignments. Also teach *Statistics 2* to 300 BSc students a year.
- Talks at IAAE 2026, CFE-CMStatistics 2026, NESG 2024 & 2026, GIFT 2025, Econometric Society Winter Meeting 2025; co-authors at NBER-NSF 2026, $(EC)^2$ 2026, Chicago Fed 2025, Purdue 2024, Erasmus 2024.

General Manager Jun 2016 – Jul 2020
NEVA Torg (retail) St. Petersburg, RU

- Took the company from break-even in 2017 to RUB 2.5m (≈ EUR 35k) after-tax profit in 2018 (∼9% net margin).
- Redesigned the stores and moved staff pay to a revenue share: revenue per store rose 30% in 2018.
- Led a team of 10, opened three stores and renegotiated supplier contracts (cost of goods down 15% on average).
- Ran the books and the financial analysis; shut a loss-making store once the numbers showed it was beyond saving.

Selected Projects

Targeted Local Projections paper code with O. Boldea · working paper, 2026
Combines local projections and VARs, trading a little bias for ∼20% shorter long-horizon intervals with no loss of coverage.

FOMC Responses to Credit Spreads with J. R. Campbell · in progress
Proposes adding credit spreads to the Fed's policy rule; in SVAR simulations, this cuts unemployment volatility by 30%.

Corrected Local Projections for Structural IRFs with O. Boldea, A. Cornea-Madeira, A. R. Hall · in progress
An estimator that removes predictable noise from local projections; theory and simulations show it has lower variance.

DSGE Identification and Testing with J. R. Campbell · in progress
A new utility function and a simpler model structure make a macro model estimable and testable directly from data.

Education

PhD, Econometrics, *Tilburg University* 2023 – 2027 (expected)
Research Master, Economics, *Tilburg University · CentER Scholarship* 2021 – 2023
MSc Economics, Monetary Policy & Banking, *University of Amsterdam · Amsterdam Merit Scholarship* 2020 – 2021
MA (Hons) Economics, *University of Glasgow* 2012 – 2016

Skills

Time series	Forecasting (VARs, out-of-sample evaluation), ARMA, state-space models, Kalman filter
Statistics	Bayesian inference (own MCMC sampler in Python), MLE, bootstrap, Monte Carlo simulation
Programming	Python (NumPy, SciPy, statsmodels, matplotlib), R, MATLAB, SQL, Stata, LaTeX, Git
Causal inference	Difference-in-differences, IV, GMM, double ML, structural VARs, local projections, impulse responses
Machine learning	Ridge, lasso, cross-validation, PCA, classification, random forests, boosting
AI tools	Claude Code and Codex for research code, simulation pipelines and document automation
Domain	Macro-finance, monetary policy, credit spreads, FRED data, DSGE and macro–climate models
Communication	Data visualisation, research papers and reports, teaching, conference talks
Business	Financial statements, accounting, business analytics, supplier negotiation, team leadership
Languages	English (C2), Dutch (B1), Russian (native)